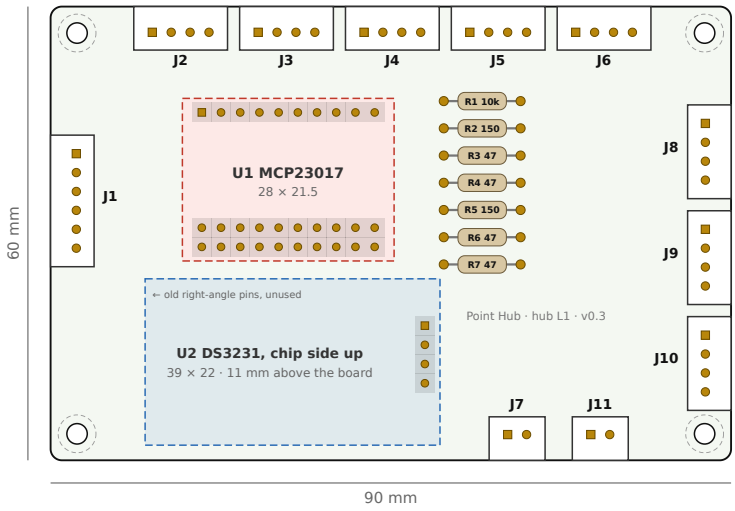
Hub board, Level 1 — paper check at 1:1

Point Hub board · v0.3 · 13.11.2026 · seen from the parts side · print at 100 %



50 mm: check with the ruler before you lay parts on it

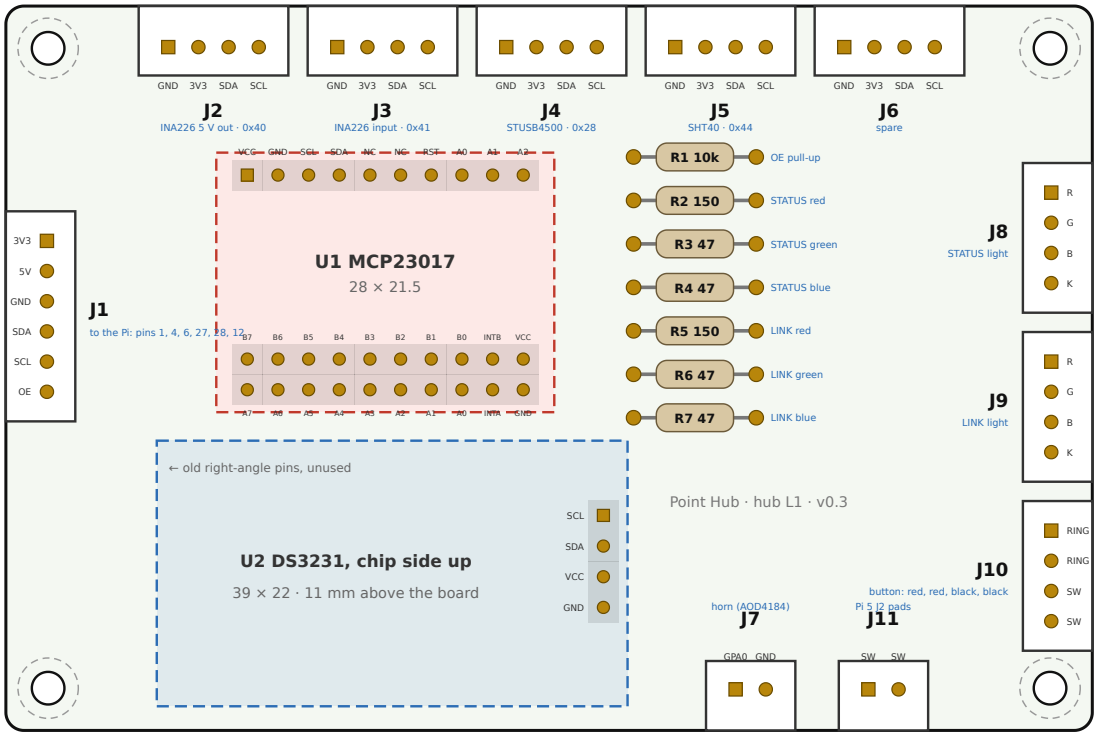
What to check

1. Print page 1 at **100 % / actual size** (not “fit to page”). The 50 mm bar must measure 50 mm, the board 90 × 60.
2. **MCP23017**, label side up: single row at the top, VCC at the left (checked 13.11). Beep test 13.11: single-row GND beeps only with the right end of the **outer** row, so the outer row is port A, as drawn.
3. **DS3231**, chip side up: the four plain holes at its right end (SCL SDA VCC GND) sit over U2, SCL on the square (checked 13.11). Cell side up is mirrored: SCL would land on GND.
4. Put the **INA226**, **AOD4184** and **button** next to the paper where they will sit in the box: are the J-sockets on the right side for their wires?
5. Tell me what moves; I make the KiCad board from the corrected positions.

Squares = pin 1. Red outline = MCP23017 (plugs in from above). Blue outline = DS3231 (on 4 straight pins, chip side up, cell underneath). Dashed circles = M2.5 holes, keep clear. The INA226 boards, AOD4184, PD board and button are **not** on the hub; they connect with cables.

Pin names (enlarged, not for measuring)

Socket pin order as seen from the parts side; pin 1 is the square pad.



Notes

1. Everything runs at **3.3 V** (MCP23017, DS3231, INA226, SHT40). The only 5 V on the hub is the button ring.
2. I²C pull-ups come from the modules. With all of them on the bus they add up: if the bus is unreliable, cut the I²C pull-up jumper on the STUSB4500 board first.
3. The I²C ports are JST-XH, not Qwiic: through-hole, easy to solder by hand, and they lock. Wire each pigtail to the module's own pin labels (the INA226 order is VCC SDA SCL ALE GND).
4. The J1 lead: XH-6 on the hub, six single Dupont housings on the Pi's pins 1, 4, 6, 12, 27, 28 through the tall stacking header.
5. DS3231: solder 4 straight male pins into its plain SCL SDA VCC GND holes, plastic on the cell side, pins pointing down (with the Pinecil, in the same session as removing the module's charging resistor “201” next to D2). Leave the right-angle header on; put tape over its pins so they touch nothing.
6. Positions follow ruler photos (±1 mm). Nothing is ordered until the paper check is done.

Connections

Net	Joins
3V3	J1·1 (Pi pin 1) → every port J2–J6 pin 2, U1 VCC (both), U1 RST, U2 VCC, R1
5V	J1·2 (Pi pin 4) → J10·1 (button ring)
GND	J1·3 (Pi pin 6) → ports J2–J6 pin 1, U1 GND (both), U1 address pins A0 A1 A2 (single row), U2 GND, J7·2, J8·4, J9·4, J10·2
SDA	J1·4 (Pi pin 27, GPIO0) → ports J2–J6 pin 3, U1 SDA, U2 SDA
SCL	J1·5 (Pi pin 28, GPIO1) → ports J2–J6 pin 4, U1 SCL, U2 SCL
OE	J1·6 (Pi pin 12, GPIO18) → R1 10k → 3V3: panels stay dark at power-on
Horn	U1 GPA0 (port-A row “A0”) → J7·1 (horn: AOD4184 PWM; J7·2 → its GND)
STATUS	U1 GPA1 → R2 150 → J8·1 red; GPA2 → R3 47 → J8·2 green; GPA3 → R4 47 → J8·3 blue; J8·4 = common cathode
LINK	U1 GPA4 → R5 150 → J9·1 red; GPA5 → R6 47 → J9·2 green; GPA6 → R7 47 → J9·3 blue; J9·4 = common cathode
Button	J10·3 → J11·1, J10·4 → J11·2 (straight through to the Pi 5 J2 pads; either way round)
Unused	U1 port B, INTA, INTB, GPA7 (input); the DS3231 right-angle header (32K, SQW and a second copy of the bus)

Ports

Socket	Pins	Goes to
J1	3V3 · 5V · GND · SDA · SCL · OE	to the Pi: pins 1, 4, 6, 27, 28, 12
J2	GND · 3V3 · SDA · SCL	INA226 5 V out · 0x40
J3	GND · 3V3 · SDA · SCL	INA226 input · 0x41
J4	GND · 3V3 · SDA · SCL	STUSB4500 · 0x28
J5	GND · 3V3 · SDA · SCL	SHT40 · 0x44
J6	GND · 3V3 · SDA · SCL	spare
J7	GPA0 · GND	horn (AOD4184)
J8	R · G · B · K	STATUS light
J9	R · G · B · K	LINK light
J10	RING · RING · SW · SW	button: red, red, black, black
J11	SW · SW	Pi 5 J2 pads

Parts for the hub

Part	How many	State
Hub PCB	90 × 60, 2 layers, 1.6 mm, JLCPCB (5 pieces is the minimum order)	after this check, in KiCad
JST-XH sockets, top entry, through-hole	2-pin × 2, 4-pin × 8, 6-pin × 1	to buy
JST-XH pigtails, pre-crimped, 20–30 cm	2-pin × 2, 4-pin × 8, 6-pin × 1 (no crimping needed)	to buy
Female headers 2.54 mm, 8.5 mm tall	1 × 10 (×1), 2 × 10 (×1), 1 × 4 (×1)	to buy (or from a kit)
Male pin header 2.54 mm, straight	1 × 4 for the DS3231 (the MCP23017 bag has its own)	kit
Dupont female housings 1-pin	6, for the Pi end of the J1 lead	to buy (or cut a jumper)
Resistors	10k × 1, 150 × 2, 47 × 4	arrived 12.11
M2.5 standoffs + screws	4	kit